Deriving the law of universal gravitation in the weak field case of GR

Arin Kharkar - University of Washington

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1 Introduction

Newton's law of universal gravitation has long provided a highly accurate model for gravity, including being used for putting a man on the moon. Observations such as the recession of Mercury's perihelion however, have shown that for objects moving where $v << c$ does not hold, or near objects of great mass, deviations from newton's law occur. In this document, we calculate the acceleration an object feels from a massive spherically symmetric non-rotating body in GR and show how when certain conditions are applied, we can recover Newton's law of gravitation.

2 Review of concepts

2.1 Newtons Law of Universal Gravitation

Through observation, Isaac Newton discovered that the gravitational force between two objects is proportional to the product of their masses over the square of their separation distance, or that

$$F_g = G\frac{m_1 m_2}{r^2} \tag{1}$$

Since in GR, forces are rather complicated to work with, we simplify this equation into

$$a = G\frac{M}{r^2} \tag{2}$$

Where a is the acceleration a small object feels from a large object of mass M at a distance of separation r.

2.2 The Shwarzschild Solution

The Schwarzschild metric describes the spacetime around a spherically symmetric non-rotating body, given by:

$$g_{\mu\nu} = \mathrm{diag}(-(1-\frac{2GM}{rc^2})c^2, (1-\frac{2GM}{rc^2})^{-1}, r^2, r^2\sin^2\theta) \tag{3}$$

Where we are using the 3D polar coordinate system:

$$x^\mu = (t, r, \theta, \phi) \tag{4}$$

Where t is coordinate time, r is the distance from the center of the body, θ is the polar angle, and ϕ is the azimuthal angle.

We will also be using the mostly plus (-,+,+,+) convention.

2.3 The Geodesic Equation

The geodesic equation:

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\alpha\beta}\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau} = 0 \tag{5}$$

Describes how objects follow spacetime curvature. where τ is the proper time along the particle's worldline. The Christoffel symbols, which encode the effects of spacetime curvature, are defined by:

$$\Gamma^\mu_{\alpha\beta} = \frac{1}{2}g^{\mu\lambda}(\frac{\partial g_{\lambda\alpha}}{\partial x^\beta} + \frac{\partial g_{\lambda\beta}}{\partial x^\alpha} - \frac{\partial g_{\alpha\beta}}{\partial x^\lambda}) \tag{6}$$

where repeated indices are summed over (Einstein summation convention) and $g^{\mu\nu}$ is the inverse metric satisfying $g^{\mu\nu}g_{\nu\rho} = \delta^\mu_\rho$.

3 The Derivation

In order to show that $a = \frac{GM}{r^2}$ in the weak field case, we want to calculate $\frac{d^2x^r}{d\tau^2}$ where $\frac{dx^\mu}{d\tau} = 0$ for $\mu \in (\theta, \phi)$ since we want to consider the path of a particle falling directly towards the body with no angular deflection.

3.1 The metric & Christoffel Symbols

We begin with restating both the definition of the Christoffel symbols and the metric we are using (Schwarzschild)

$$\Gamma^\mu_{\alpha\beta} = \frac{1}{2}g^{\mu\lambda}(\frac{\partial g_{\lambda\alpha}}{\partial x^\beta} + \frac{\partial g_{\lambda\beta}}{\partial x^\alpha} - \frac{\partial g_{\alpha\beta}}{\partial x^\lambda}) \tag{7}$$

$$g_{\mu\nu} = \text{diag}(-(1-\frac{2M}{r}), (1-\frac{2M}{r})^{-1}, r^2, r^2\sin^2\theta) \tag{8}$$

Since the Schwarzschild metric is purely diagonal, that is $g_{\mu\nu} = 0$ if $\mu \neq \nu$, we can simplify the definition of the Christoffel Symbols to

$$\Gamma^\mu_{\alpha\beta} = \frac{1}{2}g^{\mu\mu}(\frac{\partial g_{\mu\alpha}}{\partial x^\beta} + \frac{\partial g_{\mu\beta}}{\partial x^\alpha} - \frac{\partial g_{\alpha\beta}}{\partial x^\mu}) \tag{9}$$

Since when $\lambda \neq \mu$, $g_{\mu\nu} = 0$.

To compute $\frac{d^2x^r}{d\tau^2}$ we solve the geodesic equation

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\alpha\beta}\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau} = 0 \tag{10}$$

for $\mu = r$

We only need to sum for $\alpha, \beta \in (r, t)$ since for (θ, ϕ) $\frac{dx^\mu}{d\tau} = 0$

$$\frac{d^2r}{d\tau^2} + \Gamma^r_{rr}(\frac{dr}{d\tau})^2 + \Gamma^r_{tt}(\frac{dt}{d\tau})^2 = 0 \tag{11}$$

3.2 Computing the Christoffel Symbols

To solve this, we must compute the christoffel symbols $\Gamma^r_{\mu\nu}$:

$$\Gamma^r_{rr} = \frac{1}{2}g^{rr}(\frac{\partial g_{rr}}{\partial x^r}) \tag{12}$$

$$= \frac{1}{2}(1 - \frac{2M}{r})\frac{2M}{r^2} \tag{13}$$

$$\boxed{\Gamma^r_{rr} = (1 - \frac{2M}{r})\frac{M}{r^2}} \tag{14}$$

$$\Gamma^r_{rt} = \frac{1}{2}g^{rr}(\frac{\partial g_{rr}}{\partial x^t}) = 0 \tag{15}$$

$$\Gamma^r_{tr} = \frac{1}{2}g^{rr}(\frac{\partial g_{rr}}{\partial x^t}) = 0 \tag{16}$$

$$\Gamma^r_{tt} = \frac{1}{2}g^{rr}(\frac{\partial g_{rr}}{\partial x^t}) \tag{17}$$

$$\boxed{\Gamma^r_{tt} = -\frac{1}{2}(1 - \frac{2M}{r})(\frac{2M}{r^2})} \tag{18}$$

3.3 Putting it all Together

Now lets plug in these values and solve!

$$\frac{d^2x^r}{d\tau^2} = -\Gamma^r_{rr}(\frac{dr}{d\tau})^2 + \Gamma^r_{tt}(\frac{dt}{d\tau})^2 \tag{19}$$

$$\frac{d^2r}{d\tau^2} = -(1 - \frac{2M}{r})\frac{M}{r^2}(\frac{dr}{d\tau})^2 - \frac{1}{2}(1 - \frac{2M}{r})(\frac{2M}{r^2})(\frac{dt}{d\tau})^2 \tag{20}$$

Now lets re-add units using dimensional analysis:

$$\frac{d^2r}{d\tau^2} = -\frac{GM}{r^2c^2}(\frac{dr}{d\tau})^2 + \frac{2G^2M^2}{r^3c^4}(\frac{dr}{d\tau})^2 + \frac{GM}{r^2}(\frac{dt}{d\tau})^2 - \frac{2G^2M^2}{r^3c^2}(\frac{dt}{d\tau})^2 \tag{21}$$

Since when $v << c$ and in weaker fields, $\frac{dt}{d\tau} \approx 1$ (Time dilation is not very apparent)

$$\frac{d^2r}{d\tau^2} = -\frac{GM}{r^2c^2}(\frac{dr}{d\tau})^2 + \frac{2G^2M^2}{r^3c^4}(\frac{dr}{d\tau})^2 + \frac{GM}{r^2} - \frac{2G^2M^2}{r^3c^2} \tag{22}$$

Rearranging in terms of rough magnitude:

$$\frac{d^2r}{d\tau^2} = \frac{GM}{r^2} - \frac{GM}{r^2c^2}(\frac{dr}{d\tau})^2 - \frac{2G^2M^2}{r^3c^2} + \frac{2G^2M^2}{r^3c^4}(\frac{dr}{d\tau})^2 \tag{23}$$

It is clear that the first term, $\frac{GM}{r^2}$ is much larger than the others, which are many orders of magnitude smaller, having terms of c^{-2} and G^2 or higher, so we can thus conclude that in a weak field and when $\frac{dr}{d\tau} << c$:

$$\frac{d^2r}{d\tau^2} = \frac{GM}{r^2} \tag{24}$$

Exactly what Newton's laws predict!

4 Conclusion

In the case mentioned earlier, that of Mercury's perihelion, the tiny deviation was measurable simply because of Mercury's closeness to the Sun; the correction terms began to play a non-negligible (but still tiny, one arcsecond per century) factor.